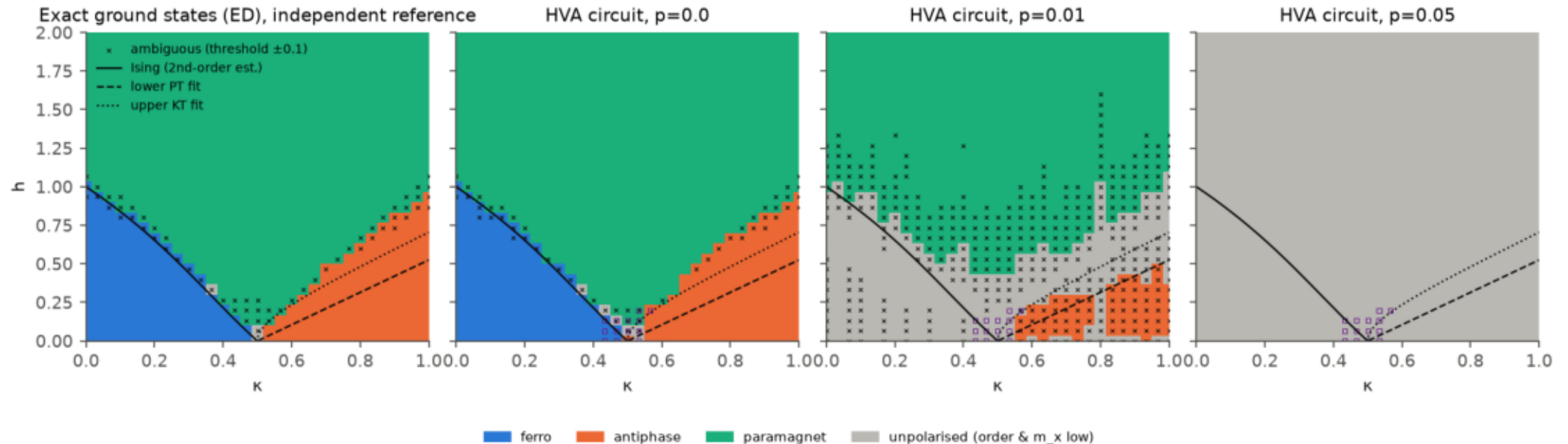


ANNNI phase diagram under CNOT-target noise

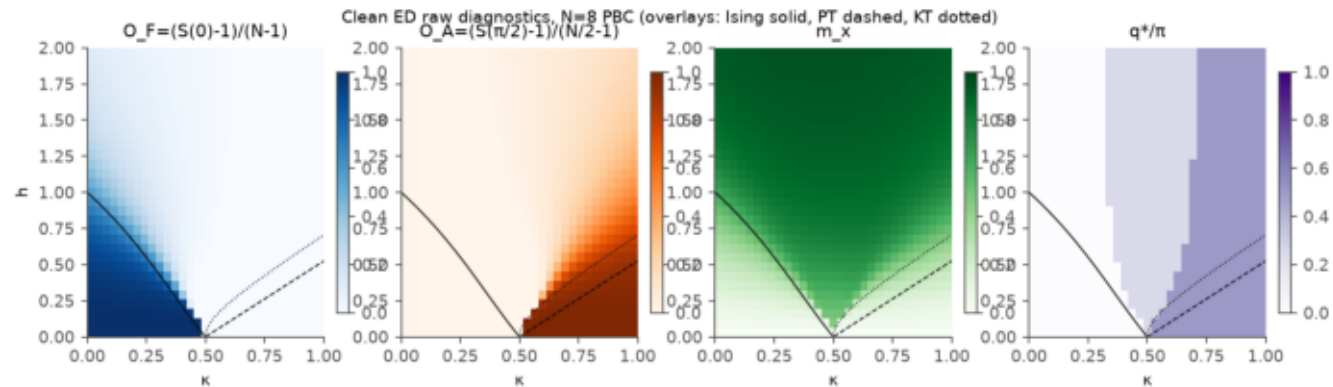
Benjamin · Dimitri · Yan · Frederick

- Exact reference (sparse ED, $N = 8, 10, 12$) and a symmetry-preserving variational circuit ($N = 8, 128$ CNOTs)
- Same circuit evaluated at $p = 0, 0.01, 0.05$ with `DepolarizingChannel(p)` on every CNOT target (default.mixed)
- Headline: $p = 0.01$ erases the ferromagnet under fixed thresholds, yet derivative-defined boundaries barely move;
 - $p = 0.05$ leaves an essentially maximally mixed state; ZNE recovers the $p = 0.01$ map (98 % label agreement)
- Floating phase: compatible signatures at $N = 12$, not resolved



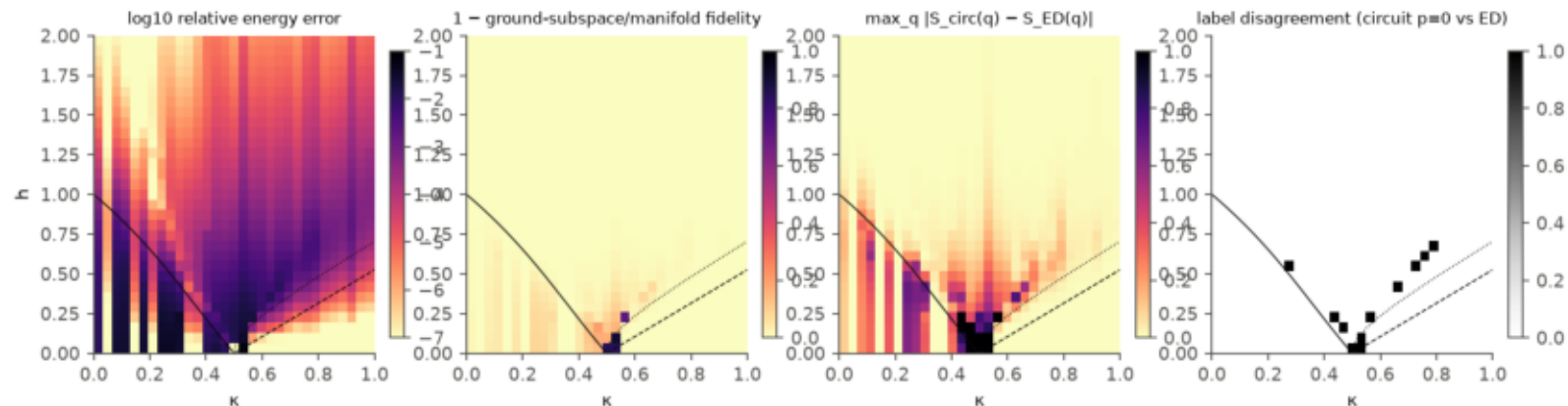
Model, diagnostics and references

- $H = -\sum Z_i Z_{i+1} + \kappa \sum Z_i Z_{i+2} - h \sum X_i$, Pauli ops, periodic ring
- Finite symmetric ground states have $\langle Z_i \rangle = 0 \rightarrow$ use correlations
- $S(q) = (1/N) \sum_{ij} e^{iq(i-j)} \langle Z_i Z_j \rangle$ ($C_{ii} = 1$ kept)
 - $O_F = (S(0)-1)/(N-1)$, $O_A = (S(\pi/2)-1)/(N/2-1)$, m_x
 - checks: ferro $S(0)=N$, antiphase $S(\pi/2)=N/2$, $|+\rangle$ and $I/2^N$ both $S=1$
- Overlays are approximate thermodynamic curves:
 - Ising 2nd-order estimate ($h_I(0)=1$), lower PT fit, upper KT fit
- Thresholds calibrated on limiting-case anchors, not on the curves



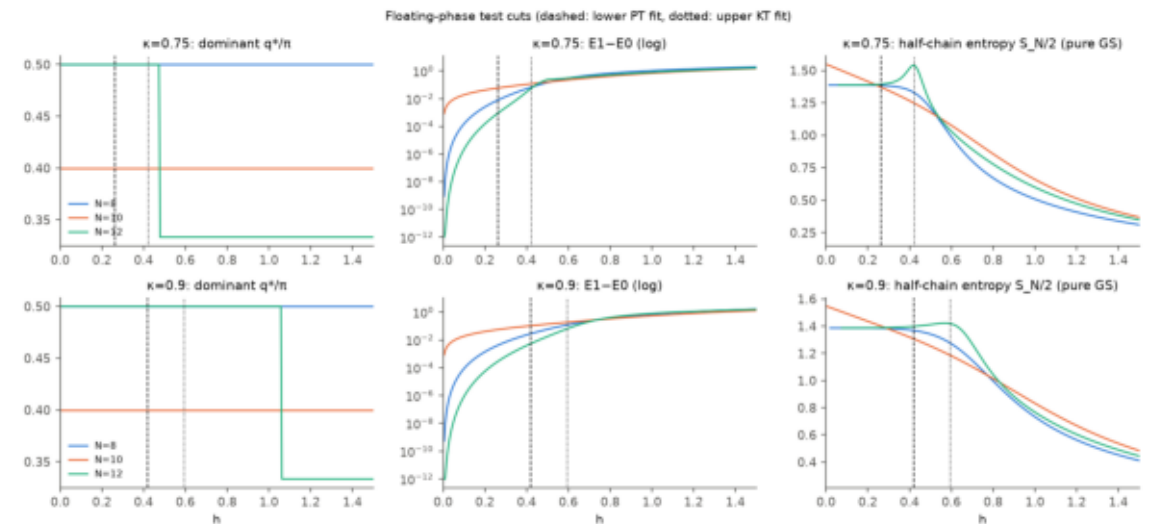
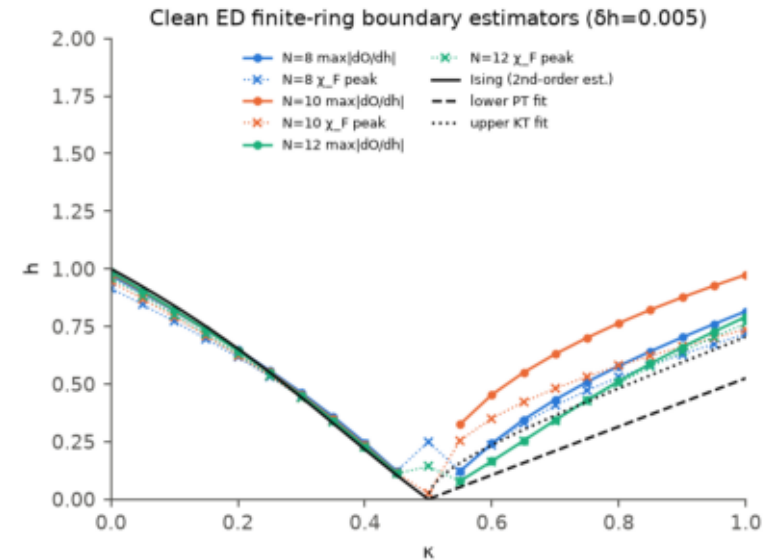
Method and validation

- Sparse eigsh, $k = 6$, warm + independent cold start ($\Delta E < 1e-13$)
- HVA: $|+\rangle$ then $4 \times [ZZ_{nn}, ZZ_{nnn}, X]$ rotations, 12 parameters
 - RY/CNOT ring stalled at 1–3 % energy error $\rightarrow$ switched
 - adjoint-gradient L-BFGS, forward + reverse warm sweeps
- Clean circuit vs ED: median relE $4e-5$, median 1–F $5e-5$, labels 98.9 %
- Noise: 128 target channels; tape check; NumPy DM engine = default.mixed to $3e-14$
- Sharded, resumable CLIs; every figure regenerated from .npz



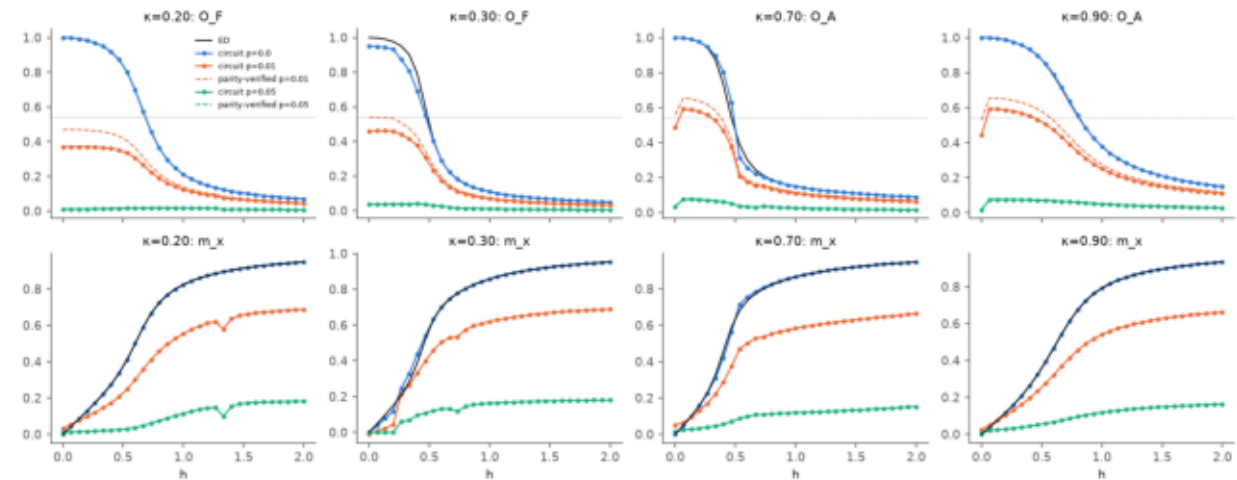
Clean phase diagram and finite size

- Ferro-para boundary $\approx$ N-independent, tracks the Ising estimate
 - $\kappa=0.2$: χ_F peak 0.616 (N=8) $\rightarrow$ 0.629 (N=12); DMRG 0.639
- Antiphase boundary drifts down with N toward the KT fit
 - $\kappa=0.75$: 0.562 (N=8) $\rightarrow$ 0.432 (N=12); KT fit 0.423
- N = 10 cannot host period 4: commensurability anomaly
- Floating: N=12 entropy peak near KT line, q^* locked at $\pi/2 \rightarrow$ unresolved



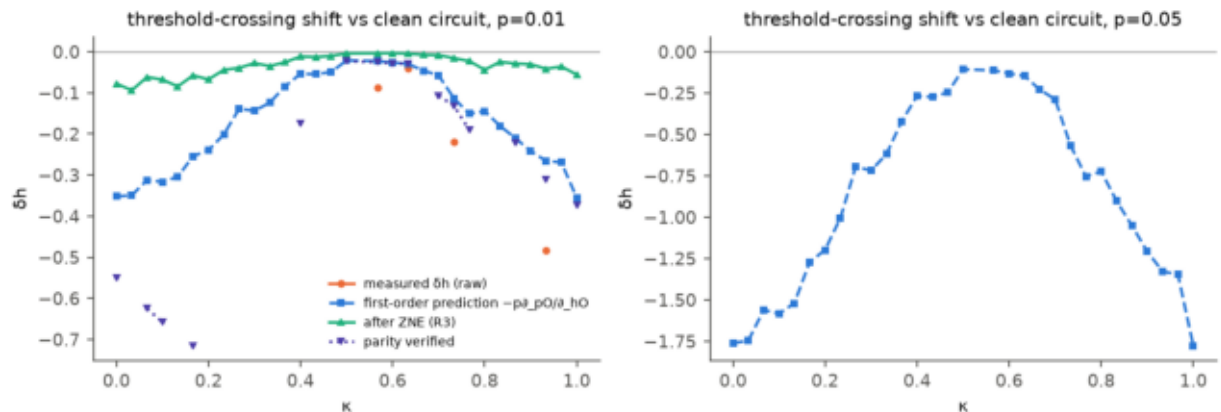
Noise: what shifts, by how much

- $p=0.01$: order contracts to 0.60 (median) of its clean value
 - ferro $O_F(h \rightarrow 0)$ $1.00 \rightarrow 0.37$; antiphase O_A $1.00 \rightarrow 0.59$
 - fixed threshold: ferro 0 % kept, antiphase 46 %, para 89 %
 - surviving threshold crossings move to lower h ($-0.04 \dots -0.48$)
 - derivative peaks: median $|\delta h|$ 0.005 (28/31 rows ≤ 0.03)
- $p=0.05$: purity 0.0047 vs 0.0039 maximally mixed; all 'unpolarised'
- Noise does not change H : these are prepared-state diagnostics



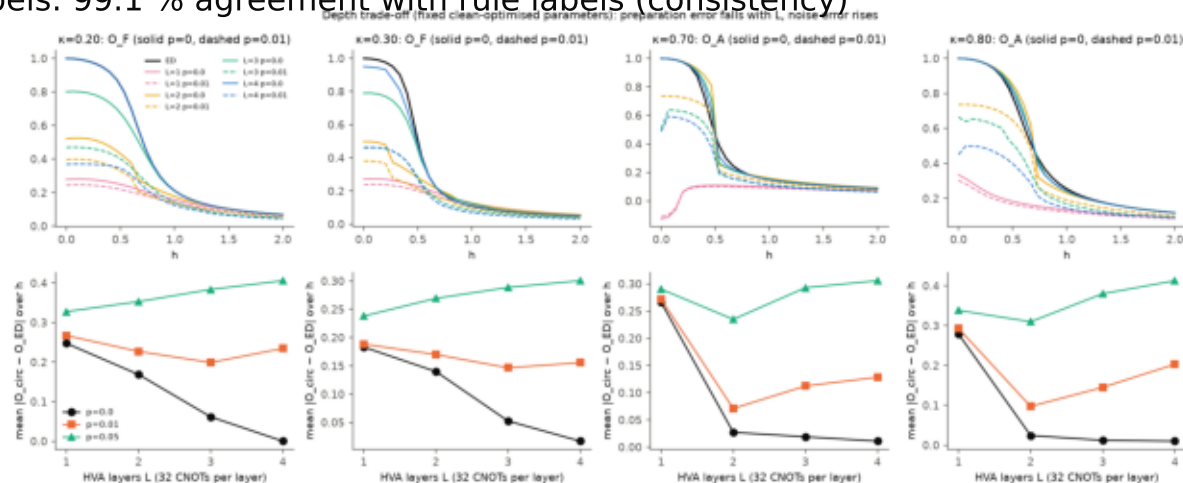
Why: exact channel-insertion response

- $d\langle O \rangle / dp = \sum_g \text{Tr}\{O_g L_t(\rho_g)\}$ over all 128 channel locations [S28]
- $d \ln O / dp$: ferro -99 , antiphase -69 , m_x (para) $-32 \dots -48$
 - terminal-noise equivalent (16 hits/site): -43 for pair correlators
- Mid-circuit errors are carried by later CNOT-RZ-CNOT gadgets
- Early layers contribute most (ferro: -27 layer 1 vs -17 layer 4)
- First-order $-p \partial_p O / \partial_h O$ has the right sign; measured shifts $1.5\text{--}4\times$ larger (nonlinear)
- Robustness ranking (ferro weakest) is circuit- and diagnostic-specific



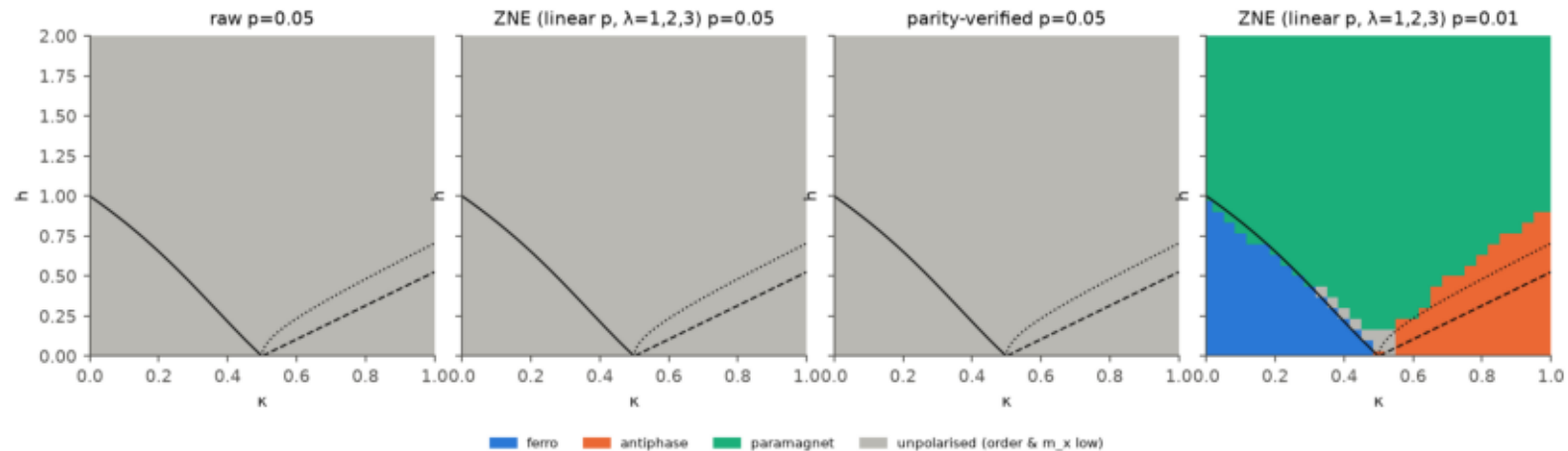
Depth, re-optimisation, second classifier

- Same experiment for HVA depth $L = 1 \dots 4$ (32 CNOTs per layer) on four κ cuts
 - best L at $p=0.01$: 3 (ferro), 2 (antiphase); at $p=0.05$: 1-2
- Antiphase is accurate already at $L=2$ (error 0.024); ferro needs $L=4$
 - the GHZ-like ferro state uses the full depth, so it carries the most channels
- Noise-aware re-optimisation (min $\text{Tr } H p_p$): energy drops 0.2–1.0, order moves ≤ 0.03
 - optimiser buys energy via m_x by leaving the ground state: energy-optimal $\neq$ diagnostic-optimal
- k-means on $(S(q)/N, m_x)$, no labels: 99.1 % agreement with rule labels (consistency)



Error mitigation

- ZNE: Richardson on each C_{ij} and $\langle X_i \rangle$ at λp , $\lambda = 1, 2, 3$; $p=0$ held out
 - $p=0.01$: $|\Delta C_{ij}|$ 0.063 $\rightarrow$ 0.011, labels 70 % $\rightarrow$ 98 %, $|\delta h| \leq 0.09$
 - cost: variance $\times 19$; $p=0.05$: fails (signal ~ 1 %)
- Parity verification $P = \prod X_i$: acceptance 71 % ($p=0.01$)
 - $|\Delta C_{ij}|$ 0.063 $\rightarrow$ 0.053: most errors stay in the even sector
- Mitigation cannot fix finite-N physics or ansatz bias



Take-aways and limitations

- Accurate clean diagram on 8–12-site rings; ferro side converged, antiphase side drifting to KT
- For a circuit good enough to represent these ground states (128 CNOTs):
 - $p = 0.01$ removes absolute order (ferro first) while boundary positions stay put
 - $p = 0.05$ leaves no usable signal; depth vs noise is the real trade-off
- ZNE turns $p = 0.01$ back into a correct diagram; parity checks help little
- Quench dynamics (Trotter from $|+\rangle$): Loschmidt-rate peaks only for quenches deep into the ordered side ($h \leq 0.4$); channel error at $p=0.01$ (≈ 0.28) exceeds the Trotter error (≤ 0.035)
- Limits: $N \leq 12$, approximate overlays, $\kappa \approx 0.5$ masked, fixed-parameter protocol only, no shot noise
- All code, data, notebook: [Scientific Track/team/](#)